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Axial Encoder C-Pulse Set Up Issue

christopher hardiman (christopher c hardiman), 06/18/2002 - 08:32 pm



PROPRIETARY WARNING

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Communique Number

MQUE-5B2NBV

[Revision #CH-E87F of **Axial Encoder C-Pulse Set Up Issue** by **Christopher C Hardiman**]

Equipment

LightSpeed Plus & Ultra

Problem

Images are slightly rotated from one to the next. Home flag position and C-Pulse coincidence is not occurring at the proper point. Home flag is shifted from center either too high in the detector window, or too low, with respect to the C-Pulse. FE attempting to adjust the C-Pulse position is unable to do so per the manual.

Solution

Service documentation is adequate for setting the encoder-to-home flag relationship. The issue of rotated images is manifested when the encoder manufacturer fails to set the C-pulse-to-drive gear relationship as indicated in 2328307ADW (see attachment). If this relationship is not maintained as indicated, then the C-Pulse could, worst case, land gear tooth, to gear tooth. That is, the gantry drive gear teeth are in contact with the encoder drive gear teeth when the coincidence is established. Once shifted over to mesh the two gears, the C-pulse will shift. This creates an error in the coincidence between the home flag and the C-pulse leading to a rotated shift in the images. The field does not have the required tooling and measuring devices to set the gear properly. Setting the C-pulse manually is, at best, difficult given the location of the LED and the Encoder. Field sites experiencing the problem should order and replace the encoder assemblies.

You can download any of the files below by dragging them to your Windows desktop. To open a file for reading, double-click it.



Author

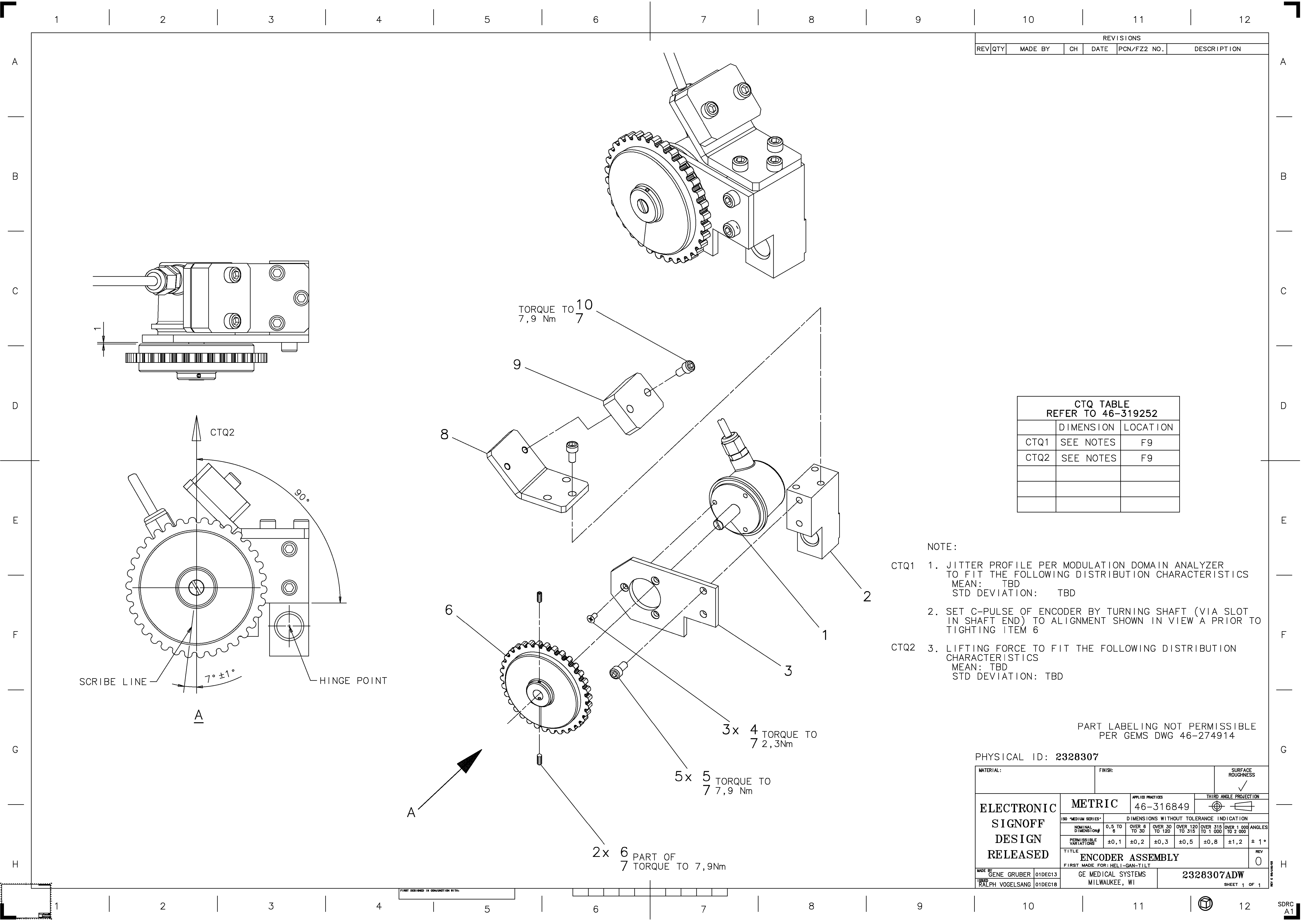
Christopher C Hardiman

Revision

06/18/2002 08:32 PM

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REVISIONS					
REV	QTY	MADE BY	CH	DATE	PCN/FZ2 NO.
DESCRIPTION					

CTQ TABLE REFER TO 46-319252		
	DIMENSION	LOCATION
CTQ1	SEE NOTES	F9
CTQ2	SEE NOTES	F9

- NOTE:
- CTQ1 1. JITTER PROFILE PER MODULATION DOMAIN ANALYZER TO FIT THE FOLLOWING DISTRIBUTION CHARACTERISTICS
MEAN: TBD
STD DEVIATION: TBD
2. SET C-PULSE OF ENCODER BY TURNING SHAFT (VIA SLOT IN SHAFT END) TO ALIGNMENT SHOWN IN VIEW A PRIOR TO TIGHTING ITEM 6
- CTQ2 3. LIFTING FORCE TO FIT THE FOLLOWING DISTRIBUTION CHARACTERISTICS
MEAN: TBD
STD DEVIATION: TBD

PART LABELING NOT PERMISSIBLE
PER GEMS DWG 46-274914

PHYSICAL ID: 2328307

MATERIAL:		FINISH:		SURFACE ROUGHNESS	
				✓	
ELECTRONIC SIGNOFF DESIGN RELEASED		METRIC 46-316849		THIRD ANGLE PROJECTION	
ISO "MEDIUM SERIES"		DIMENSIONS WITHOUT TOLERANCE INDICATION			
NOMINAL DIMENSION#		0,5 TO 6	OVER 6 TO 30	OVER 30 TO 120	OVER 120 TO 315
PERMISSIBLE VARIATIONS		±0,1	±0,2	±0,3	±0,5
TITLE		ENCODER ASSEMBLY	REV		0
FIRST MADE FOR: HELI-GAN-TILT					
MADE BY GENE GRUBER 01DEC13		GE MEDICAL SYSTEMS MILWAUKEE, WI		2328307ADW	
ISSUED RALPH VOGELSANG 01DEC18				SHEET 1 OF 1	